Hybrid Video Provenance Framework (HVPF)

Phase 3 Results: Vision Transformer, Ensembles and PRNU Injection into AI Video

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Abstract

Phase 3 adds a Vision Transformer on I-frame noise residuals, calibrated models, model ensembles and equal-length inputs for the sequence models. It also adds a harder attack, taken from Khan et al. (2025): a real camera's PRNU injected into AI-generated video, so that the clip passes conventional PRNU verification. As in Phase 2, the task is a three-class decision for each video clip (organic / manipulated / synthetic), compared with the conventional PRNU method (PCE rule). All numbers are generated directly from the result files.

1 Key results

- **Best result:** Stacking (LR) reaches macro F1 0.851, PVA 0.836 and authentic-vs-not AUC 0.941 (EER 0.149) in 5-fold cross-validation on camera models never seen in training (Phase 2 best: XGBoost, F1 0.792). The conventional PCE rule reaches AUC 0.549 (RQ3); it dropped from Phase 2 because injected AI clips pass it.
- **Vision Transformer:** the best single model (F1 0.820) and by far the most robust on unseen sources: it recognises 90% of FloreView genuine videos (XGBoost 55%), including stabilised phones without a usable PRNU (Table 9). Plain averaging of all models (F1 0.814) does not beat the ViT alone; stacking, which learns how much to trust each model, does (0.851).
- **Calibration is not worth it:** per-class threshold offsets change macro F1 on the test cameras by +0.004 (XGBoost) but cost PVA on unseen sources (XGBoost -5, ViT -4 points). Stacking, also fitted on validation data, is the best combination on both (Table 5).
- **PRNU injected into AI video (Khan-style):** 2,984 AI clips carry a real camera's PRNU. At full strength the PCE rule accepts 98% of them as genuine footage of that camera (96% for unseen generators); stacking accepts 0% (1%) (Table 8).
- **Weakest spots:** (1) AI clips from unseen generators that were also WhatsApp-like compressed: stacking calls 42% synthetic, 42% manipulated and 16% organic. (2) Genuine VISION videos that went through our own re-encoding: stacking recognises 45% of them (15% of the WhatsApp-like ones); almost all others are called manipulated, because the attacked clips went through the same encoder (Table 7).
- **10 frames:** with equal-length inputs, 10 frames is slightly better than 5 even on length-matched clips (+1 to +2 PVA points); combinations of the Step 8 models at 10 frames reach F1 0.844 without the ViT, so a 10-frame ViT and stack is the next likely gain.
- **Features (RQ1):** removing Noiseprint statistics costs the most (about 0.05 macro F1), followed by TSNCS and PRNU (0.01–0.03); FFT and residual moments add almost nothing (Figure 2).

2 What changed in Phase 3

- **Vision Transformer (Step 8b).** Pretrained ViT-S/16 (21.7M parameters, ImageNet-21k) fine-tuned on three SRM high-pass residuals of each I-frame (random 224×224 crops and flips, AdamW, mixed precision, class-balanced loss, early stopping on validation macro F1). A clip's prediction averages its first five I-frames × four crops. This is the same input size as PDC-ViT.
- **Calibration and ensembles (Step 8c).** Validation predictions of every model are now kept. *+cal*: three class offsets on the log-probabilities tuned for validation macro F1. *Ensembles*: mean class probability of SVM+XGB, of the three sequence models, of all learned models, and of the three best models on validation data; *stacking*: logistic regression on all models' validation outputs. Validation data always comes from other camera models than the test data.
- **Equal-length inputs.** Clips shorter than the window repeat their own frames, so sequence models cannot use the number of I-frames as a shortcut (a Phase 2 limitation).
- **Wider tuning.** XGBoost: depth 4/6/8 and a min-child-weight variant, up to 1500 trees; SVM: C up to 30.
- **Harder synthetic class.** Khan-style AI clips with an injected camera PRNU ($\alpha = 0.5$ and 1, same strength units as the manipulation attacks) are added to training and test. The attacker estimates the fingerprint from the camera's own natural videos, never from the analyst's reference videos. Cameras without a usable video PRNU are not impersonated.

3 Data and protocol

The evaluation uses 27,583 clips. Model inputs come from the first five I-frames of each clip. Results are averaged over five grouped cross-validation folds: no camera model is in both training and test. *Realistic* uses all clips; *encoder-controlled* uses only clips encoded by our libx264 pipeline (genuine re-encodes, attacks, AI clips), so encoder traces cannot separate the classes. *Cross-dataset* is FloreView (genuine) plus GenBuster (unseen AI generators), tested with the fixed-split models.

Table 1: Clips per class and source.

Class	Source	Clips
AI + injected camera PRNU	GenBuster	1,200
AI + injected camera PRNU	GenVidBench	1,784
manipulated	VISION-attack	7,200
organic	ACID	4,380
organic	ACID-attack	3,168
organic	FloreView	1,716
organic	VISION	1,259
organic	VISION-attack	2,400
synthetic	GenBuster	1,800
synthetic	GenVidBench	2,676

4 Main results

Table 2: Realistic setup, 5-fold cross-validation (mean ± std over folds), 5 frames. PVA = share of clips assigned to their true provenance class; AUC/EER for genuine vs not genuine. Lower block: calibrated models and combinations (Step 8c).

Model	PVA	Macro F1	AUC (3-class)	AUC genuine vs not	EER
PCE rule	–	–	–	0.549	0.449
SVM	0.765 ± 0.025	0.769 ± 0.025	0.898	0.884	0.201
XGBoost	0.791 ± 0.030	0.794 ± 0.029	0.927	0.913	0.177
GRU	0.773 ± 0.027	0.779 ± 0.022	0.919	0.906	0.190
1D-CNN	0.768 ± 0.029	0.774 ± 0.022	0.910	0.895	0.197
Transformer	0.770 ± 0.027	0.780 ± 0.024	0.916	0.902	0.194
ViT (residual)	0.806 ± 0.020	0.820 ± 0.017	0.933	0.912	0.182
Ens. SVM+XGB	0.789 ± 0.025	0.794 ± 0.022	0.926	0.912	0.179
Ens. GRU+CNN+Transf.	0.784 ± 0.027	0.793 ± 0.022	0.924	0.912	0.182
Ens. all	0.807 ± 0.026	0.814 ± 0.022	0.941	0.927	0.167
Ens. top-3 (val)	0.818 ± 0.025	0.827 ± 0.019	0.946	0.932	0.161
Stacking (LR)	0.836 ± 0.019	0.851 ± 0.014	0.955	0.941	0.149
XGBoost + cal.	0.787 ± 0.032	0.797 ± 0.028	0.927	0.913	0.178
Ens. all + cal.	0.804 ± 0.027	0.814 ± 0.020	0.940	0.927	0.168
Ens. top-3 (val) + cal.	0.812 ± 0.026	0.826 ± 0.019	0.946	0.932	0.162

Table 3: Encoder-controlled setup (5-fold) and cross-dataset test (FloreView + GenBuster), 5 frames.

	Encoder-controlled			Cross-dataset		
Model	PVA (%)	Macro F1	AUC	PVA (%)	Macro F1	AUC
PCE rule	–	–	0.513	–	–	0.406
SVM	74	0.751	0.860	56	0.630	0.793
XGBoost	76	0.768	0.886	64	0.728	0.922
GRU	75	0.756	0.876	53	0.619	0.825
1D-CNN	75	0.758	0.872	60	0.660	0.844
Transformer	75	0.756	0.872	60	0.715	0.889
ViT (residual)	75	0.763	0.856	82	0.842	0.942
Ens. SVM+XGB	76	0.770	0.886	60	0.667	0.884
Ens. GRU+CNN+Transf.	76	0.769	0.884	59	0.683	0.866
Ens. all	78	0.785	0.898	64	0.722	0.913
Ens. top-3 (val)	78	0.789	0.897	70	0.776	0.944
Stacking (LR)	81	0.816	0.906	82	0.863	0.965
XGBoost + cal.	76	0.774	0.887	59	0.698	0.919
Ens. all + cal.	78	0.792	0.899	60	0.703	0.904
Ens. top-3 (val) + cal.	78	0.792	0.897	67	0.759	0.943

Table 4: Phase 2 vs Phase 3 (realistic, 5-fold, 5 frames). Phase 3 also contains the injected AI clips, so its test set is larger and harder.

	PVA (%)		Macro F1		AUC genuine vs not	
Model	Phase 2	Phase 3	Phase 2	Phase 3	Phase 2	Phase 3
PCE rule	–	–	–	–	0.635	0.549
XGBoost	78	79	0.792	0.794	0.899	0.913
GRU	77	77	0.785	0.779	0.891	0.906
Transformer	76	77	0.775	0.780	0.880	0.902
Best model	78	84	0.792	0.851	0.899	0.941

5 Calibration and ensembles on unseen sources

Calibration moves the decision thresholds to where they work best on the validation cameras. That barely helps on test cameras from the same datasets, and FloreView phones and unseen AI generators look different, so the moved thresholds hurt there. Stacking is also fitted on validation data, but it learns how much weight each model gets (the ViT, the most robust model on unseen sources, is one of them) and is the best combination on both. Fixed split, 5 frames:

Table 5: Effect of calibration and combination: same datasets vs unseen sources.

	Test (VISION, ACID, GenVidBench)		Cross-dataset	
Model	Macro F1	PVA (%)	PVA (%)	AUC
XGBoost	0.806	81	64	0.922
XGBoost + cal.	0.811	81	59	0.919
Ens. all	0.824	82	64	0.913
Ens. all + cal.	0.831	83	60	0.904
Ens. top-3 (val)	0.841	84	70	0.944
Ens. top-3 (val) + cal.	0.844	84	67	0.943
Stacking (LR)	0.866	86	82	0.965

6 5 vs 10 frames

The length-matched columns use only clips with at least 10 I-frames (2,520 test clips per fold), so the clip length carries no information; since Phase 3 the sequence models get equal-length inputs on all clips as well. Unlike Phase 2, 10 frames is now slightly better than 5 even when lengths are matched: more frames give a better noise estimate. The larger gain on all clips comes from the long clips (VISION and attacks), whose 10-frame features are more reliable. The ViT and the main results use 5 frames, because ACID and most AI clips have only about six I-frames.

Table 6: Effect of the frame window (realistic, 5-fold).

	All clips				Length-matched ($\geq$ 10 frames)			
	5 frames		10 frames		5 frames		10 frames	
Model	PVA	AUC	PVA	AUC	PVA	AUC	PVA	AUC
XGBoost	79	0.913	82	0.927	72	0.833	74	0.848
GRU	77	0.906	81	0.924	70	0.812	71	0.824
1D-CNN	77	0.895	81	0.922	68	0.788	70	0.819
Transformer	77	0.902	81	0.919	68	0.793	70	0.819

7 Results per condition (RQ2)

Share of clips assigned to their true class (%), realistic setup, 5-fold, 5 frames. Values below 70% are in bold. Ens. = mean probability of all learned models; Stack = Stacking (LR). The PCE rule only decides genuine vs not genuine: its high rates on attacks and AI clips come with rejecting most genuine clips.

Table 7: Detection rate per condition.

Condition	n	PCE	XGB	GRU	ViT	Ens.	Stack
Genuine (organic)							
ACID native	4380	36	94	93	98	95	97
VISION native	425	39	87	84	83	88	88
VISION real WhatsApp copy	423	5	**60**	**39**	**49**	**48**	**48**
VISION real YouTube copy	411	10	**68**	**65**	73	**68**	**69**
ACID re-encode, CRF 18	528	28	92	89	84	93	93
ACID re-encode, CRF 20	528	25	90	86	83	91	89
ACID re-encode, CRF 20 + WA-like	528	0	**64**	**33**	**56**	**48**	**50**
ACID re-encode, CRF 20 + YT-like	528	8	87	81	88	86	86
ACID re-encode, CRF 23	528	20	88	83	87	89	89
ACID re-encode, CRF 28	528	9	87	86	92	90	90
VISION re-encode, CRF 18	400	24	74	70	**55**	74	**64**
VISION re-encode, CRF 20	400	21	**64**	**61**	**49**	**60**	**49**
VISION re-encode, CRF 20 + WA-like	400	2	**34**	**22**	**20**	**22**	**15**
VISION re-encode, CRF 20 + YT-like	400	10	**48**	**44**	**57**	**46**	**43**
VISION re-encode, CRF 23	400	17	**61**	**54**	**55**	**56**	**48**
VISION re-encode, CRF 28	400	9	**53**	**52**	**61**	**54**	**53**
Manipulated							
denoise_a1	400	84	**58**	**62**	77	**67**	80
denoise_a2	400	84	70	74	85	78	88
denoise_a2 + WA-like	400	99	**69**	76	81	78	88
denoise_a2 + YT-like	400	95	**65**	72	**64**	73	77
denoise_a4	400	88	80	82	91	85	93
injection_a0.25	400	83	**65**	**62**	**62**	**68**	72
injection_a0.5	400	30	73	70	72	76	82
injection_a0.5 + WA-like	400	98	72	77	76	79	86
injection_a0.5 + YT-like	400	74	**67**	**69**	**53**	72	72
injection_a1	400	2	77	78	84	83	93
removal_a0.25	400	87	**43**	**46**	**62**	**50**	**63**
removal_a0.5	400	96	75	72	72	76	88
removal_a0.5 + WA-like	400	100	**68**	73	76	76	85
removal_a0.5 + YT-like	400	98	**67**	**67**	**52**	70	74
removal_a1	400	100	93	90	84	94	98
removal_opt	400	99	**61**	**63**	**65**	**66**	76
removal_opt + WA-like	400	100	70	73	76	76	84
removal_opt + YT-like	400	100	**59**	**60**	**49**	**62**	**68**
AI-generated (synthetic)							
CogVideo	300	100	98	99	99	99	99
CogVideo + WA-like	300	100	97	97	100	98	99
CogVideo + YT-like	300	100	97	95	100	97	100
Kling	269	100	88	91	100	93	99
Kling + WA-like	269	100	86	87	99	95	98
Kling + YT-like	269	100	84	86	99	91	99
OpenSora	272	100	99	93	98	97	98
OpenSora + WA-like	272	100	100	99	98	100	99
OpenSora + YT-like	272	100	100	99	99	99	99
Sora	51	100	**35**	**49**	88	**49**	80
Sora + WA-like	51	100	**43**	**39**	**55**	**47**	**57**

Condition	n	PCE	XGB	GRU	ViT	Ens.	Stack
Sora + YT-like	51	100	**47**	**53**	88	**57**	73
AI-generated + injected camera PRNU							
CogVideo, injected α=0.5	300	13	95	96	99	96	100
CogVideo, injected α=1	300	2	96	94	99	95	100
Kling, injected α=0.5	269	17	84	90	100	93	99
Kling, injected α=1	269	2	87	86	99	88	99
OpenSora, injected α=0.5	272	16	97	96	99	98	100
OpenSora, injected α=1	272	1	97	94	99	94	99
Sora, injected α=0.5	51	27	**35**	**45**	90	**49**	88
Sora, injected α=1	51	6	**61**	**61**	92	75	84

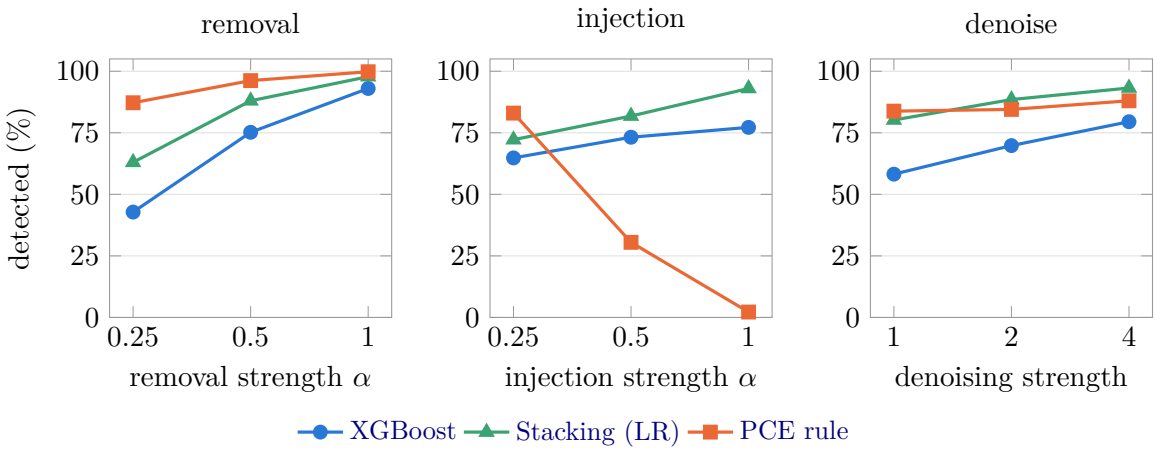


Figure 1: Attack strength vs detection (VISION attacks, realistic, 5-fold). The PCE rule "detects" removal only because low PCE is its sign of non-genuine; it misses injection, where the victim's fingerprint is matched.

8 PRNU injected into AI-generated video

Khan et al. (2025) transfer a camera's PRNU into AI-generated *images* so that they pass source verification. We apply the same attack to video: the attacker estimates the fingerprint of a real camera from that camera's own videos and adds it to every frame of an AI clip, which is then encoded like all our clips. Table 8 gives the share of clips accepted as genuine (PCE rule: PCE_0 with the claimed camera above 60; models: predicted *organic*). GenVidBench clips are from the 5-fold test folds; GenBuster generators were never seen in training.

Table 8: AI clips accepted as genuine camera footage (%, lower is better for the defender).

	GenVidBench (seen generators)			GenBuster (unseen generators)		
Detector	plain	α=0.5	α=1	plain	α=0.5	α=1
Khan et al. 2025 [2], AI images	85.5 (images, average)					
PCE rule	0	84	98	0	77	96
SVM	5	4	2	6	6	3
XGBoost	5	2	0	6	3	1
GRU	3	2	0	5	3	0
1D-CNN	5	2	0	10	4	0
Transformer	6	3	1	6	4	1
ViT (residual)	1	1	0	15	9	4
Ens. all	3	2	0	4	3	0
Stacking (LR)	1	0	0	1	2	1

Injecting a camera's PRNU does not hide the AI origin: of the injected GenVidBench clips, Stacking (LR) calls 99% synthetic, 1% manipulated and 0% organic. The injection makes the clip match the camera for the PCE rule, but it does not remove the other traces the models use (Noiseprint statistics, residual and temporal patterns).

9 Cross-dataset test

FloreView cameras are split by their PRNU baseline: 10 of 45 cameras have a usable video PRNU (attribution $\geq 50\%$); the others are modern phones whose stabilisation removes it.

Table 9: Cross-dataset detection rate (%), models trained on VISION + ACID + GenVidBench (fixed split).

Condition	n	PCE	XGB	GRU	ViT	Ens.	Stack
FloreView genuine, stabilised (no PRNU)	1361	0	49	33	88	45	73
FloreView genuine, usable PRNU	355	62	78	57	96	76	91
GenBuster AI	600	100	73	62	84	76	97
GenBuster AI + WA-like	600	100	34	32	37	40	42
GenBuster AI + YT-like	600	100	75	53	78	70	91
GenBuster AI + injected PRNU, α=0.5	600	23	79	79	90	84	97
GenBuster AI + injected PRNU, α=1	600	4	86	86	95	88	97

10 Feature ablation (RQ1)

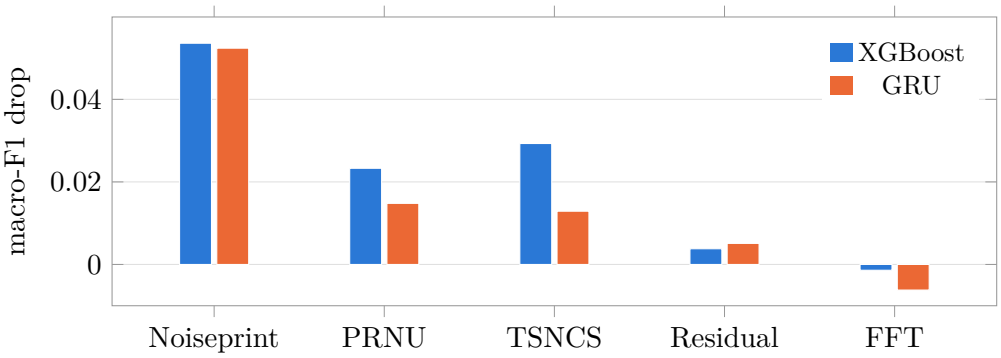


Figure 2: Drop in macro F1 when one feature group is removed (realistic, 5-fold, 5 frames). Larger = more important.

Table 10: Macro-F1 drop per feature group and model (5-fold, 5 frames).

Setup	Removed group	SVM	XGB	GRU	CNN	Transf.
realistic	Noiseprint	0.053	0.054	0.052	0.048	0.067
realistic	PRNU	0.006	0.023	0.015	0.005	0.014
realistic	TSNCS	0.016	0.029	0.013	0.015	0.013
realistic	Residual	0.001	0.004	0.005	-0.001	0.004
realistic	FFT	0.001	-0.001	-0.006	0.003	0.005
controlled	Noiseprint	0.054	0.046	0.047	0.062	0.054
controlled	PRNU	0.014	0.025	0.006	0.019	0.011
controlled	TSNCS	0.013	0.027	0.011	0.019	0.011
controlled	Residual	0.000	0.001	0.003	0.006	0.014
controlled	FFT	-0.001	0.003	-0.003	0.003	0.004

11 Comparison with the base papers

The base papers solve related but different tasks, so each is compared on its own terms. Literature values are taken from the papers as reported; protocols differ (noted per table).

11.1 Camera identification: PDC-ViT (Elharrouss et al. 2025)

PDC-ViT is a learned camera-identification model (80/20 train-test split, every camera seen in training). The closest component of HVPF is the training-free PRNU baseline (Step 4): fingerprints from flat videos, tested on natural videos, closed-set over all cameras of the dataset; verification at $PCE_0 > 60$. Our ViT shares PDC-ViT's input size but answers a different question (provenance class, not camera identity), on camera models it never saw in training.

Table 11: Camera identification. Accuracy = closed-set attribution; FNR/FPR = verification.

Method	Dataset (cameras)	Accuracy (%)	FNR (%)	FPR (%)	EER (%)
PDC-ViT [1]	VISION (35)	94.30	–	–	–
HVPF PRNU	VISION (35)	73.4	36.2	0.00	18.3
HVPF PRNU	VISION (21 non-stabilised)	100.0	–	–	–
PRNU [1]	Video-ACID + WA (26)	–	41.9	3.6	–
WA-PRNU [1]	Video-ACID + WA (26)	–	28.06	2.8	–
PDC-ViT [1]	Video-ACID + WA (26)	74.77	23.4	0.92	–
HVPF PRNU	Video-ACID (44)	59.1	59.6	0.00	25.5
HVPF PRNU	FloreView (45)	26.2	80.8	0.00	38.8

PRNU matching needs no training and is perfect on non-stabilised VISION cameras, above PDC-ViT; electronic stabilisation is what lowers the all-camera figures. Our verification threshold is strict (FPR 0%), so FNR is higher than PDC-ViT's (FPR 0.92%); the EER gives a threshold-free view. PDC-ViT was not tested on manipulated or synthetic video, and no base paper uses FloreView.

11.2 PRNU-transfer attack: Khan et al. 2025

Khan et al. report that 85.5% of PRNU-transferred images pass source verification. Table 8 covers their AI-image case on video; Table 12 covers transfer between real cameras (our *injection* attack). Compromise rate = share of injected clips accepted as genuine (realistic, 5-fold, 5 frames).

Table 12: Compromise rate of PRNU injection between cameras (%, lower is better for the defender).

	Injection strength α			α=0.5, re-shared		
Detector	0.25	0.5	1	WA-like	YT-like	All
Khan et al. 2025 [2], images			–			85.5
PCE rule	17	70	98	2	26	42
SVM	28	23	8	24	27	22
XGBoost	32	18	2	24	30	22
GRU	32	21	4	18	26	20
1D-CNN	30	22	6	22	24	21
Transformer	32	20	5	14	26	19
ViT (residual)	36	26	11	21	44	27
Ens. all	28	18	2	18	26	19
Stacking (LR)	26	14	1	13	26	16

A full-strength injection fools the PCE rule in 98% of the clips, in line with Khan et al.; the best learned detector lets through only 1.2%. Weak injections (α=0.25) rarely fool PCE but are

also harder for the models to spot.

11.3 Low false-positive operation: Bao et al. 2024

Bao et al. detect frame deletion (a different manipulation) from inter-frame noise consistency, the idea behind TSNCS, and stress performance at low false-positive rates; their results are given as ROC curves only. Table 13 reports HVPF at the same kind of operating points: share of non-genuine clips detected while at most 1, 5 or 10% of genuine clips are flagged.

Table 13: Detection rate of non-genuine clips at fixed false-positive rates (realistic, 5-fold, 5 frames).

Detector	FPR 1%	FPR 5%	FPR 10%
PCE rule	5.3	9.9	16.3
SVM	34.4	55.5	65.7
XGBoost	48.2	62.3	72.0
GRU	42.7	60.9	69.9
1D-CNN	28.5	55.9	67.0
Transformer	44.5	60.4	69.8
ViT (residual)	48.6	61.3	71.6
Ens. all	52.0	66.2	75.2
Stacking (LR)	56.4	70.5	78.7

11.4 What the base papers do not cover

- PRNU injection and removal attacks on *video*, including platform-like re-sharing (Khan et al. test images only).
- PRNU transfer into AI-generated *video*, tested against unseen generators.
- Camera identification under attack (PDC-ViT reports clean data only).
- FloreView as an unseen-camera test set.
- One model that separates genuine, manipulated and AI-generated video.

12 Limitations and next steps

- Phones whose electronic stabilisation removes the PRNU (most FloreView cameras): the feature-based models still miss about half of them; the ViT recognises most. More such phones in training would help every model.
- Platform-like copies make attacks and unseen-generator AI clips much harder to see, WhatsApp-like re-encoding most of all (Tables 7 and 9). All platform copies of fakes are simulated; only genuine VISION videos have real WhatsApp/YouTube copies.
- The manipulated class comes from 400 videos of the 35 VISION cameras, and all attacks are our own implementation; ACID cameras appear only as genuine re-encodes.
- The ViT sees one I-frame at a time and was run at 5 frames; 10-frame and multi-frame versions are the natural next step.
- Sora is represented by only 51 clips; more high-quality modern generators are needed.

References

[1] O. Elharrouss et al., "PDC-ViT: source camera identification using pixel difference convolution and vision transformer," *Neural Computing and Applications*, 2025, doi:10.1007/s00521-025-11004-z.

[2] Khan, Hasan, Islam, "Spoofing camera source attribution via PRNU transfer attacks," *ISC 2025*, Springer LNCS 16186, pp. 279–299, 2025.

[3] Bao et al., "Anti-forensics video forgery detection via noise transfer matrix," *Sensors* 24(16):5341, 2024, doi:10.3390/s24165341.